ASSIGNMENT 9.5
AI ASSISTED CODING

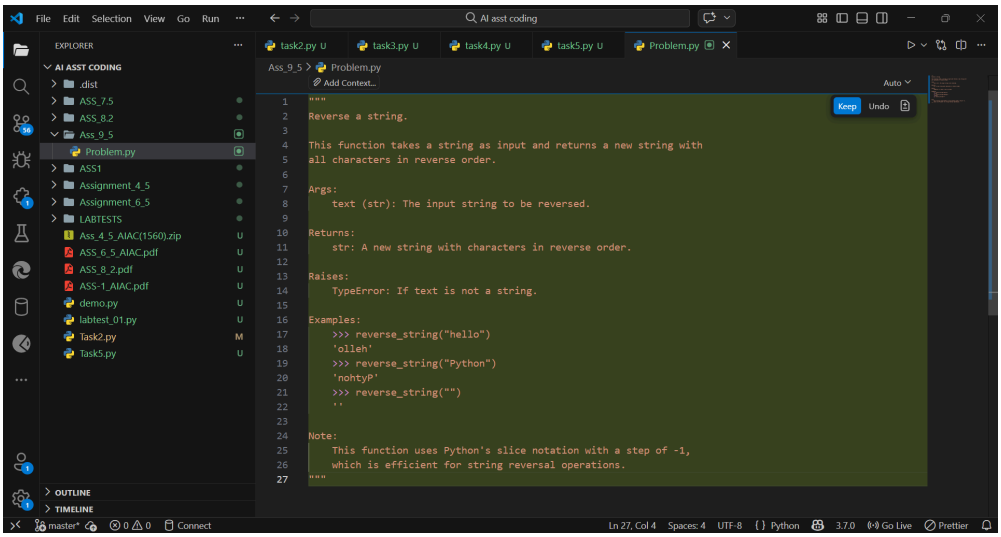
E.SAI ANURATH | BATCH-29 | 2303A51560 | DATE:20-02-2026

> Problem 1: String Utilities Function
> Consider the following Python function:
> def reverse_string(text):
> return text[::-1]
> Task:

1. Write documentation in:
   - (a) Docstring
   - (b) Inline comments
   - (c) Google-style documentation
2. Compare the three documentation styles.
3. Recommend the most suitable style for a utility-based string library.

```
"""
Reverse a string.

This function takes a string as input and returns a new string with
all characters in reverse order.

Args:
    text (str): The input string to be reversed.

Returns:
    str: A new string with characters in reverse order.

Raises:
    TypeError: If text is not a string.

Examples:
    >>> reverse_string("hello")
    'olleh'
    >>> reverse_string("Python")
    'nohtyP'
    >>> reverse_string("")
    ''

Note:
    This function uses Python's slice notation with a step of -1,
```

```
    which is efficient for string reversal operations.
"""
```

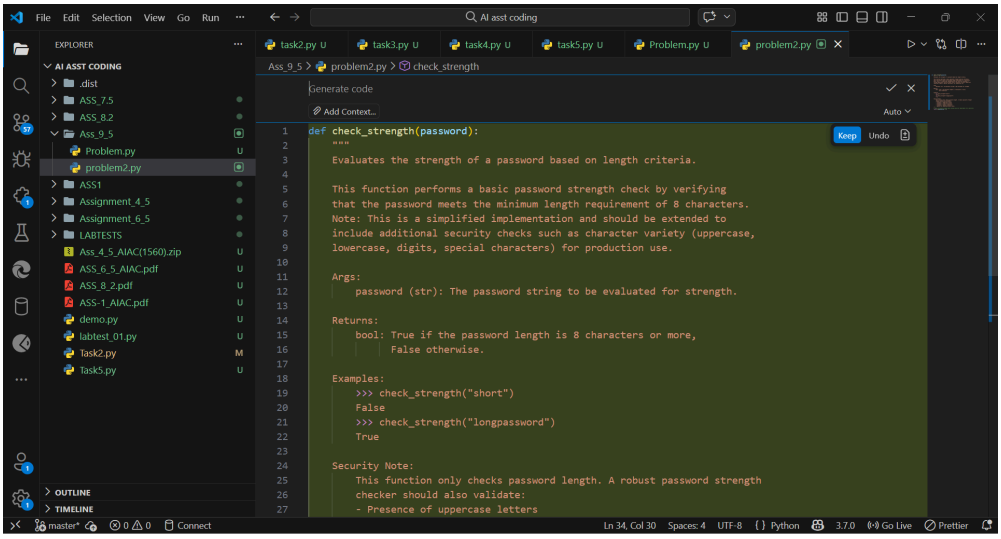


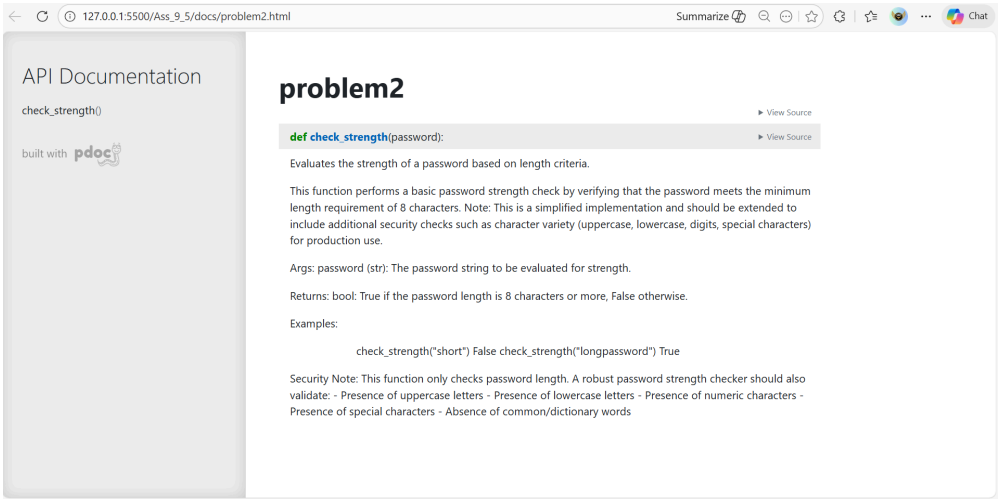
> Problem 2: Password Strength Checker
> Consider the function:
> def check_strength(password):
> return len(password) >= 8
> Task:

1. Document the function using docstring, inline comments, and Google style.
2. Compare documentation styles for security-related code.
3. Recommend the most appropriate style.

```
def check_strength(password):
    """
    Evaluates the strength of a password based on length criteria.

    This function performs a basic password strength check by verifying
    that the password meets the minimum length requirement of 8 characters.
    Note: This is a simplified implementation and should be extended to
    include additional security checks such as character variety (uppercase,
    lowercase, digits, special characters) for production use.

    Args:
        password (str): The password string to be evaluated for strength.

    Returns:
        bool: True if the password length is 8 characters or more,
              False otherwise.
```

```
    Examples:
        >>> check_strength("short")
        False
        >>> check_strength("longpassword")
        True

    Security Note:
        This function only checks password length. A robust password strength
        checker should also validate:
        - Presence of uppercase letters
        - Presence of lowercase letters
        - Presence of numeric characters
        - Presence of special characters
        - Absence of common/dictionary words
    """
    # Check if password length meets minimum security requirement of 8 characters
    return len(password) >= 8
```



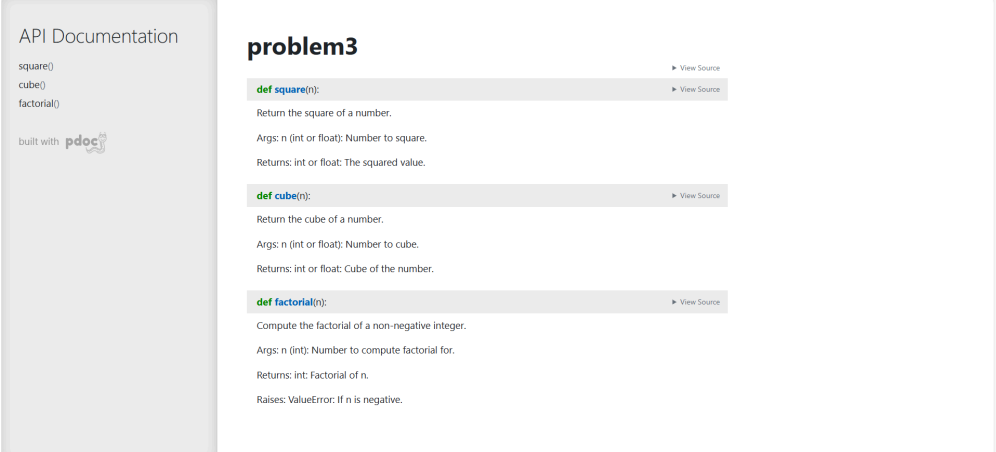


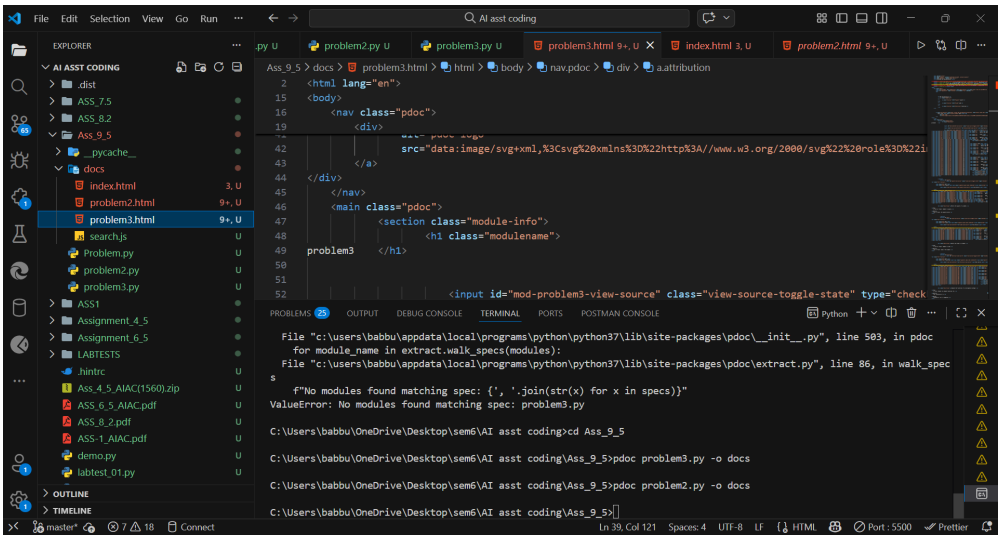
> Problem 3: Math Utilities Module
> Task:

1. Create a module math_utils.py with functions:
   o square(n)
   o cube(n)
   o factorial(n)
2. Generate docstrings automatically using AI tools.
3. Export documentation as an HTML file.

```
def square(n):
    """
    Return the square of a number.

    Args:
        n (int or float): Number to square.

    Returns:
        int or float: The squared value.
    """
    return n * n


def cube(n):
    """
    Return the cube of a number.

    Args:
        n (int or float): Number to cube.

    Returns:
        int or float: Cube of the number.
    """
    return n ** 3


def factorial(n):
    """
    Compute the factorial of a non-negative integer.

    Args:
        n (int): Number to compute factorial for.

    Returns:
        int: Factorial of n.

    Raises:
        ValueError: If n is negative.
    """
    if n < 0:
```

```
        raise ValueError("Negative numbers not allowed")
    if n == 0:
        return 1
    return n * factorial(n - 1
```





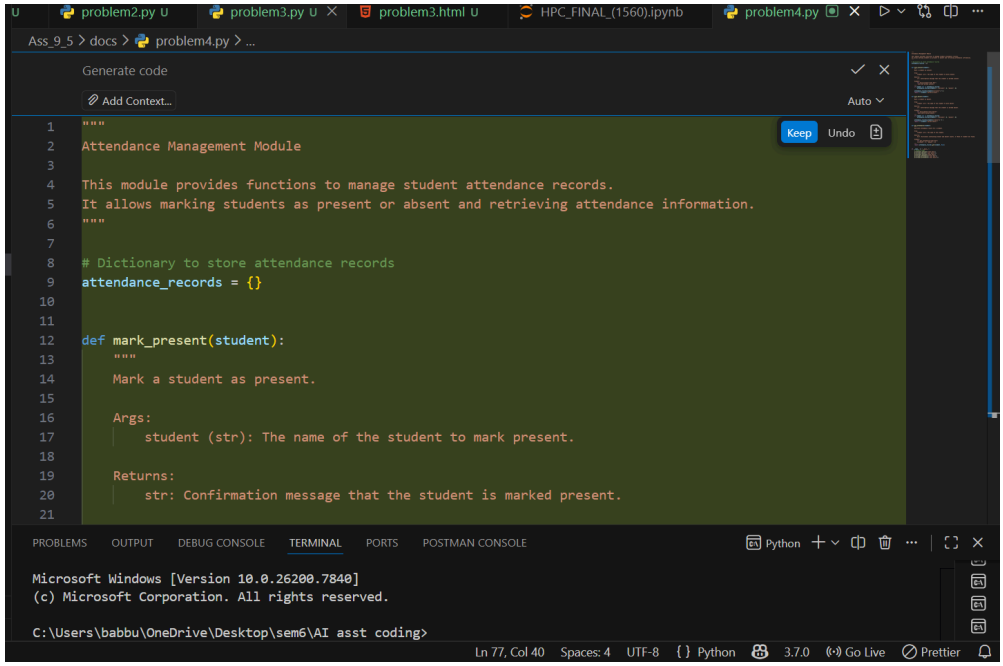
> Problem 4: Attendance Management Module
> Task:

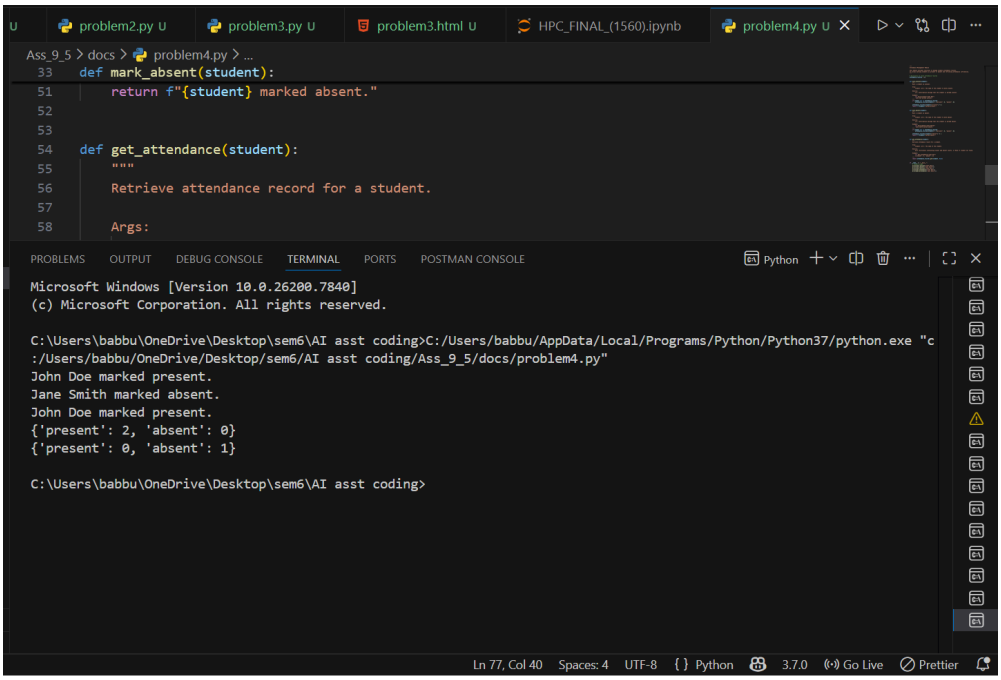
1. Create a module attendance.py with functions:
   - o mark_present(student)
   - o mark_absent(student)
   - o get_attendance(student)
2. Add proper docstrings.
3. Generate and view documentation in terminal and browse

```
"""
Attendance Management Module
```

```
This module provides functions to manage student attendance records.
It allows marking students as present or absent and retrieving attendance information.
"""

# Dictionary to store attendance records
attendance_records = {}


def mark_present(student):
    """
    Mark a student as present.

    Args:
        student (str): The name of the student to mark present.

    Returns:
        str: Confirmation message that the student is marked present.

    Example:
        >>> mark_present("John Doe")
        'John Doe marked present.'
    """
    if student not in attendance_records:
        attendance_records[student] = {"present": 0, "absent": 0}

    attendance_records[student]["present"] += 1
    return f"{student} marked present."


def mark_absent(student):
    """
    Mark a student as absent.

    Args:
        student (str): The name of the student to mark absent.

    Returns:
        str: Confirmation message that the student is marked absent.

    Example:
        >>> mark_absent("Jane Smith")
        'Jane Smith marked absent.'
    """
    if student not in attendance_records:
        attendance_records[student] = {"present": 0, "absent": 0}

    attendance_records[student]["absent"] += 1
    return f"{student} marked absent."
```

```python
def get_attendance(student):
    """
    Retrieve attendance record for a student.

    Args:
        student (str): The name of the student.

    Returns:
        dict: Dictionary containing present and absent counts, or None if student not fou
nd.

    Example:
        >>> get_attendance("John Doe")
        {'present': 5, 'absent': 2}
    """
    return attendance_records.get(student, None)


if __name__ == "__main__":
    # Example usage
    print(mark_present("John Doe"))
    print(mark_absent("Jane Smith"))
    print(mark_present("John Doe"))
    print(get_attendance("John Doe"))
    print(get_attendance("Jane Smith"))
```





> Problem 5: File Handling Function
> Consider the function:
> def read_file(filename):
> with open(filename, 'r') as f:
> return f.read()
> Task:

1. Write documentation using all three formats.
2. Identify which style best explains exception handling.
3. Justify your recommendation.

```
"""
One-line summary: Read and return the entire contents of a text file.

Extended summary:
This function opens a file in read mode and returns its complete contents as a string.
It uses a context manager (with statement) to ensure the file is properly closed
after reading, even if an exception occurs during the read operation.

Args:
    filename (str): The path to the file to be read. Can be a relative or absolute path.

Returns:
    str: The complete contents of the file as a single string.

Raises:
    FileNotFoundError: If the specified file does not exist.
    PermissionError: If the process lacks permission to read the file.
```

```
    IOError: If an error occurs while reading the file.
    IsADirectoryError: If the filename points to a directory instead of a file.

Example:
    >>> content = read_file('example.txt')
    >>> print(content)
    'Hello, World!'
"""
```

```
<!doctype html>
<html lang="en">
<head>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <meta name="generator" content="pdoc 13.1.1"/>
    <title>problem5 API documentation</title>

    <style>/*! * Bootstrap Reboot v5.0.0 (https://getbootstrap.com/) * Copyright 2011-202
1 The Bootstrap Authors * Copyright 2011-2021 Twitter, Inc. * Licensed under MIT (http
s://github.com/twbs/bootstrap/blob/main/LICENSE) * Forked from Normalize.css, licensed MI
T (https://github.com/necolas/normalize.css/blob/master/LICENSE.md) */*,::after,::before
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r:smooth}}body{margin:0;font-family:system-ui,-apple-system,"Segoe UI",Roboto,"Helvetica
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0%200%2016%2016%22%3E%3Cpath%20d%3D%22M8.982%201.566a1.13%201.13%200%200%200-1.96%200L.16
```

5%2013.233c-.457.778.091%201.767.98%201.767h13.713c.889%200%201.438-.99.98-1.767L8.982%201.566zM8%205c.535%200%20.954.462.9.995l-.35%203.507a.552.552%200%200%201-1.1%200L7.1%205.995A.905.905%200%200%201%208%205zm.002%206a1%201%200%201%201%200%202%201%201%200%200%201%200-2z%22/%3E%3C/svg%3E");}.pdoc .pdoc-alert-danger{color:#842029;background-color:#f8d7da;border-color:#f5c2c7;background-image:url("data:image/svg+xml,%3Csvg%20xmlns%3D%22http%3A//www.w3.org/2000/svg%22%20width%3D%2224%22%20height%3D%2224%22%20fill%3D%22%23842029%22%20viewBox%3D%220%200%2016%2016%22%3E%3Cpath%20d%3D%22M5.52.359A.5.5%200%200%201%206%200h4a.5.5%200%200%201%20.474.658L8.694%206H12.5a.5.5%200%200%201%20.395.807l-7%209a.5.5%200%200%201-.873-.454L6.823%209.5H3.5a.5.5%200%200%201-.48-.641l2.5-8.5z%22/%3E%3C/svg%3E");}.pdoc .visually-hidden{position:absolute !important;width:1px !important;height:1px !important;padding:0 !important;margin:-1px !important;overflow:hidden !important;clip:rect(0, 0, 0, 0) !important;white-space:nowrap !important;border:0 !important;}.pdoc h1, .pdoc h2, .pdoc h3{font-weight:300;margin:.3em 0;padding:.2em 0;}.pdoc > section:not(.module-info) h1{font-size:1.5rem;font-weight:500;}.pdoc > section:not(.module-info) h2{font-size:1.4rem;font-weight:500;}.pdoc > section:not(.module-info) h3{font-size:1.3rem;font-weight:500;}.pdoc > section:not(.module-info) h4{font-size:1.2rem;}.pdoc > section:not(.module-info) h5{font-size:1.1rem;}.pdoc a{text-decoration:none;color:var(--link);}.pdoc a:hover{color:var(--link-hover);}.pdoc blockquote{margin-left:2rem;}.pdoc pre{border-top:1px solid var(--accent2);border-bottom:1px solid var(--accent2);margin-top:0;margin-bottom:1em;padding:.5rem 0 .5rem .5rem;overflow-x:auto;background-color:var(--code);}.pdoc code{color:var(--text);padding:.2em .4em;margin:0;font-size:85%;background-color:var(--code);border-radius:6px;}.pdoc a > code{color:inherit;}.pdoc pre > code{display:inline-block;font-size:inherit;background:none;border:none;padding:0;}.pdoc > section:not(.module-info){margin-bottom:1.5rem;}.pdoc .modulename{margin-top:0;font-weight:bold;}.pdoc .modulename a{color:var(--link);transition:100ms all;}.pdoc .git-button{float:right;border:solid var(--link) 1px;}.pdoc .git-button:hover{background-color:var(--link);color:var(--pdoc-background);}.view-source-toggle-state,.view-source-toggle-state ~ .pdoc-code{display:none;}.view-source-toggle-state:checked ~ .pdoc-code{display:block;}.view-source-button{display:inline-block;float:right;font-size:.75rem;line-height:1.5rem;color:var(--muted);padding:0 .4rem 0 1.3rem;cursor:pointer;text-indent:-2px;}.view-source-button > span{visibility:hidden;}.module-info .view-source-button{float:none;display:flex;justify-content:flex-end;margin:-1.2rem .4rem -.2rem 0;}.view-source-button::before{position:absolute;content:"View Source";display:list-item;list-style-type:disclosure-closed;}.view-source-toggle-state:checked ~ .attr .view-source-button::before,.view-source-toggle-state:checked ~ .view-source-button::before{list-style-type:disclosure-open;}.pdoc .docstring{margin-bottom:1.5rem;}.pdoc section:not(.module-info) .docstring{margin-left:clamp(0rem, 5vw - 2rem, 1rem);}.pdoc .docstring .pdoc-code{margin-left:1em;margin-right:1em;}.pdoc h1:target,.pdoc h2:target,.pdoc h3:target,.pdoc h4:target,.pdoc h5:target,.pdoc h6:target,.pdoc .pdoc-code > pre > span:target{background-color:var(--active);box-shadow:-1rem 0 0 0 var(--active);}.pdoc .pdoc-code > pre > span:target{display:block;}.pdoc div:target > .attr,.pdoc section:target > .attr,.pdoc dd:target > a{background-color:var(--active);}.pdoc *{scroll-margin:2rem;}.pdoc .pdoc-code .linenos{user-select:none;}.pdoc .attr:hover{filter:contrast(0.95);}.pdoc section, .pdoc .classattr{position:relative;}.pdoc .headerlink{--width:clamp(1rem, 3vw, 2rem);position:absolute;top:0;left:calc(0rem - var(--width));transition:all 100ms ease-in-out;opacity:0;}.pdoc .headerlink::before{content:"#";display:block;text-align:center;width:var(--width);height:2.3rem;line-height:2.3rem;font-size:1.5rem;}.pdoc .attr:hover ~ .headerlink,.pdoc *:target > .headerlink,.pdoc .headerlink:hover{opacity:1;}.pdoc .attr{display:block;margin:.5rem 0 .5rem;padding:.4rem .4rem .4rem 1rem;background-color:var(--accent);overflow-x:auto;}.pdoc .classattr{margin-left:2rem;}.pdoc .name{color:var(--name);font-weight:bold;}.pdoc .def{color:var(--def);font-weight:bold;}.pdoc .signature{background-color:transparent;}.pdoc .param, .pdoc .return-annotation{white-space:pre;}.pdoc .signatu

```
re.multiline .param{display:block;}.pdoc .signature.condensed .param{display:inline-bloc
k;}.pdoc .annotation{color:var(--annotation);}.pdoc .view-value-toggle-state,.pdoc .view-
value-toggle-state ~ .default_value{display:none;}.pdoc .view-value-toggle-state:checked
~ .default_value{display:inherit;}.pdoc .view-value-button{font-size:.5rem;vertical-alig
n:middle;border-style:dashed;margin-top:-0.1rem;}.pdoc .view-value-button:hover{backgroun
d:white;}.pdoc .view-value-button::before{content:"show";text-align:center;width:2.2em;di
splay:inline-block;}.pdoc .view-value-toggle-state:checked ~ .view-value-button::before{c
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m;}.pdoc .inherited dd:not(:last-child):after{content:", ";}.pdoc .inherited .class:befor
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x;transition:all 200ms;}.pdoc table{display:block;width:max-content;max-width:100%;overfl
ow:auto;margin-bottom:1rem;}.pdoc table th{font-weight:600;}.pdoc table th, .pdoc table t
d{padding:6px 13px;border:1px solid var(--accent2);}</style>
    <style>/*! custom.css */</style></head>
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stroke-miterlimit='10' stroke-width='2' d='M4 7h22M4 15h22M4 23h22'/></svg></label>
        <input id="togglestate" type="checkbox" aria-hidden="true" tabindex="-1">
        <div>

<a class="attribution" title="pdoc: Python API documentation generator" href="https://pdo
c.dev" target="_blank">
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```

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2.36%200-1.3.52-2.34.52-1.06%201.34-1.7%201.66-1.32%203.54-1.32.76%200%201.48.22.72.2%20
1.06.4l.32.2q.36.24.56.38.52.4.52.92%200%20.5-.42%201.14-.72%201.1-1.38%201.1-.38%200-1.0
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```
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</div>
    </nav>
    <main class="pdoc">
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                    <h1 class="modulename">
problem5    </h1>

                        <div class="docstring"><p>One-line summary: Read and return the e
ntire contents of a text file.</p>

<p>Extended summary:
This function opens a file in read mode and returns its complete contents as a string.
It uses a context manager (with statement) to ensure the file is properly closed
after reading, even if an exception occurs during the read operation.</p>

<p>Args:
    filename (str): The path to the file to be read. Can be a relative or absolute path.
</p>

<p>Returns:
    str: The complete contents of the file as a single string.</p>

<p>Raises:
    FileNotFoundError: If the specified file does not exist.
    PermissionError: If the process lacks permission to read the file.
    IOError: If an error occurs while reading the file.
    IsADirectoryError: If the filename points to a directory instead of a file.</p>

<p>Example:</p>
```

```
<blockquote>
  <blockquote>
    <blockquote>
      <p>content = read_file('example.txt')
      print(content)
          'Hello, World!'</p>
    </blockquote>
  </blockquote>
</blockquote>
</div>
<input id="mod-problem5-view-source" class="view-source-toggle-state" type="checkbox" ari
a-hidden="true" tabindex="-1">

                        <label class="view-source-button" for="mod-problem5-view-source">
<span>View Source</span></label>

                        <div class="pdoc-code codehilite"><pre><span></span><span id="L-
1"><a href="#L-1"><span class="linenos"> 1</span></a><span class="sd">&quot;&quot;&quot;
</span>
</span><span id="L-2"><a href="#L-2"><span class="linenos"> 2</span></a><span class="sd">
One-line summary: Read and return the entire contents of a text file.</span>
</span><span id="L-3"><a href="#L-3"><span class="linenos"> 3</span></a>
</span><span id="L-4"><a href="#L-4"><span class="linenos"> 4</span></a><span class="sd">
Extended summary:</span>
</span><span id="L-5"><a href="#L-5"><span class="linenos"> 5</span></a><span class="sd">
This function opens a file in read mode and returns its complete contents as a string.</s
pan>
</span><span id="L-6"><a href="#L-6"><span class="linenos"> 6</span></a><span class="sd">
It uses a context manager (with statement) to ensure the file is properly closed</span>
</span><span id="L-7"><a href="#L-7"><span class="linenos"> 7</span></a><span class="sd">
after reading, even if an exception occurs during the read operation.</span>
</span><span id="L-8"><a href="#L-8"><span class="linenos"> 8</span></a>
</span><span id="L-9"><a href="#L-9"><span class="linenos"> 9</span></a><span class="sd">
Args:</span>
</span><span id="L-10"><a href="#L-10"><span class="linenos">10</span></a><span class="s
d">    filename (str): The path to the file to be read. Can be a relative or absolute pat
h.</span>
</span><span id="L-11"><a href="#L-11"><span class="linenos">11</span></a>
</span><span id="L-12"><a href="#L-12"><span class="linenos">12</span></a><span class="s
d">Returns:</span>
</span><span id="L-13"><a href="#L-13"><span class="linenos">13</span></a><span class="s
d">    str: The complete contents of the file as a single string.</span>
</span><span id="L-14"><a href="#L-14"><span class="linenos">14</span></a>
</span><span id="L-15"><a href="#L-15"><span class="linenos">15</span></a><span class="s
d">Raises:</span>
</span><span id="L-16"><a href="#L-16"><span class="linenos">16</span></a><span class="s
d">    FileNotFoundError: If the specified file does not exist.</span>
</span><span id="L-17"><a href="#L-17"><span class="linenos">17</span></a><span class="s
d">    PermissionError: If the process lacks permission to read the file.</span>
</span><span id="L-18"><a href="#L-18"><span class="linenos">18</span></a><span class="s
d">    IOError: If an error occurs while reading the file.</span>
```

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</span><span id="L-19"><a href="#L-19"><span class="linenos">19</span></a><span class="s
d">    IsADirectoryError: If the filename points to a directory instead of a file.</span>
</span><span id="L-20"><a href="#L-20"><span class="linenos">20</span></a>
</span><span id="L-21"><a href="#L-21"><span class="linenos">21</span></a><span class="s
d">Example:</span>
</span><span id="L-22"><a href="#L-22"><span class="linenos">22</span></a><span class="s
d">    &gt;&gt;&gt; content = read_file(&#39;example.txt&#39;)</span>
</span><span id="L-23"><a href="#L-23"><span class="linenos">23</span></a><span class="s
d">    &gt;&gt;&gt; print(content)</span>
</span><span id="L-24"><a href="#L-24"><span class="linenos">24</span></a><span class="s
d">    &#39;Hello, World!&#39;</span>
</span><span id="L-25"><a href="#L-25"><span class="linenos">25</span></a><span class="s
d">&quot;&quot;&quot;</span>
</span></pre></div>


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    </main>
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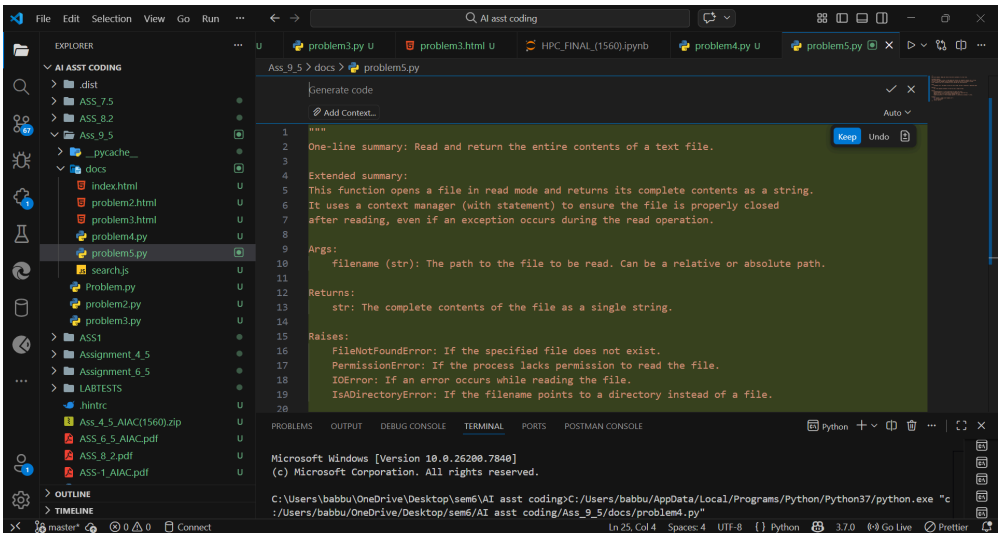

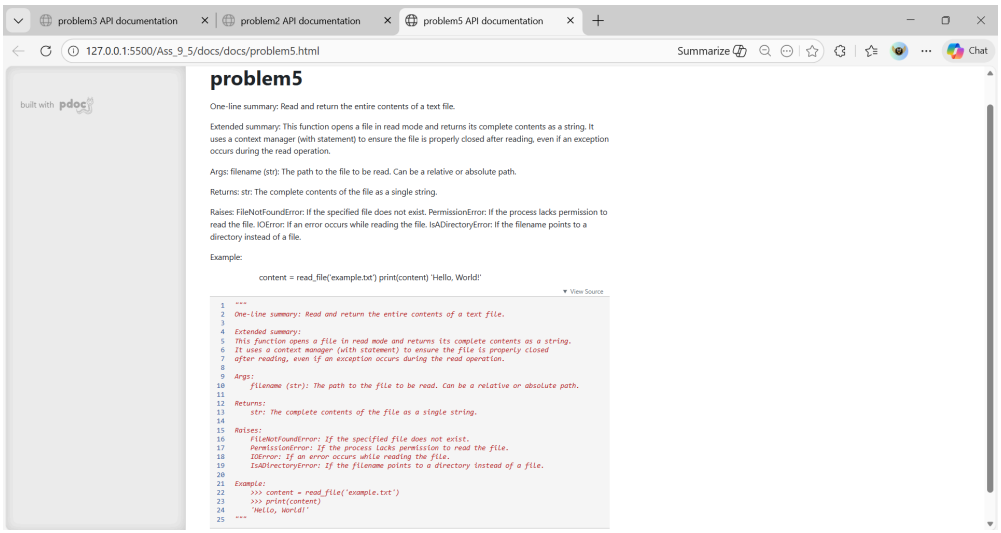
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problem2 API documentation
problem5 API documentation
127.0.0.1:5500/Ass_9_5/docs/docs/problem5.html
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One-line summary: Read and return the entire contents of a text file.
Extended summary: This function opens a file in read mode and returns its complete contents as a string. It uses a context manager (with statement) to ensure the file is properly closed after reading, even if an exception occurs during the read operation.
Args: filename (str): The path to the file to be read. Can be a relative or absolute path.
Returns: str: The complete contents of the file as a single string.
Raises: FileNotFoundError: If the specified file does not exist. PermissionError: If the process lacks permission to read the file. IOError: If an error occurs while reading the file. IsADirectoryError: If the filename points to a directory instead of a file.
Example:
content = read_file('example.txt') print(content) 'Hello, World!'
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"""
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This function opens a file in read mode and returns its complete contents as a string.
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IsADirectoryError: If the filename points to a directory instead of a file.
Example:
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>>> print(content)
'Hello, World!'
"""